MIL-T-25193B(WEPS)
27 March 1961
Superseding
MIL-T-25193A(ASG)
4 Sept 1956

MILITARY SPECIFICATION

TRANSMITTER, REMOTE COMPASS, TYPE C-2 (UNSTABILIZED)

This specification has been approved by the Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 This specification covers one type of unstabilized remote compass transmitter designated type C-2.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, and publications, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-D-70327 - Drawings, Engineering and Associated Lists
MIL-C-5541 - Chemical Films for Aluminum and Aluminum Alloys
MIL-P-7936 - Parts and Equipment, Aeronautical, Preparation for Delivery
MIL-S-6872 - Soldering Process, General Specification for
MIL-S-7742 - Screw Threads, Standard, Aeronautical
MIL-A-8625 - Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-P-116 - Preservation, method of

STANDARDS

Military

MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-130 - Identification Marking of U. S. Military Property

FSC-6605

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PUBLICATIONS

Air Force-Navy Aeronautical Bulletins

No. 143 - Specifications and Standards; Use of
No. 405 - Storage Life - Aeronautical Articles

2.2 Availability of documents - (1) When requesting any of the applicable documents refer to both title and number. Copies of this specification and applicable documents may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code CDS, 5801 Tabor Ave, Philadelphia, 20, Pennsylvania.

3. REQUIREMENTS

3.1 Preproduction - The transmitter furnished under this specification shall be a product, samples of which have been tested and passed the preproduction tests specified herein.

3.2 Component parts - This specification covers a remote compass transmitter. The compass transmitter shall consist of one compensator and one transmitter.

3.3 Materials -

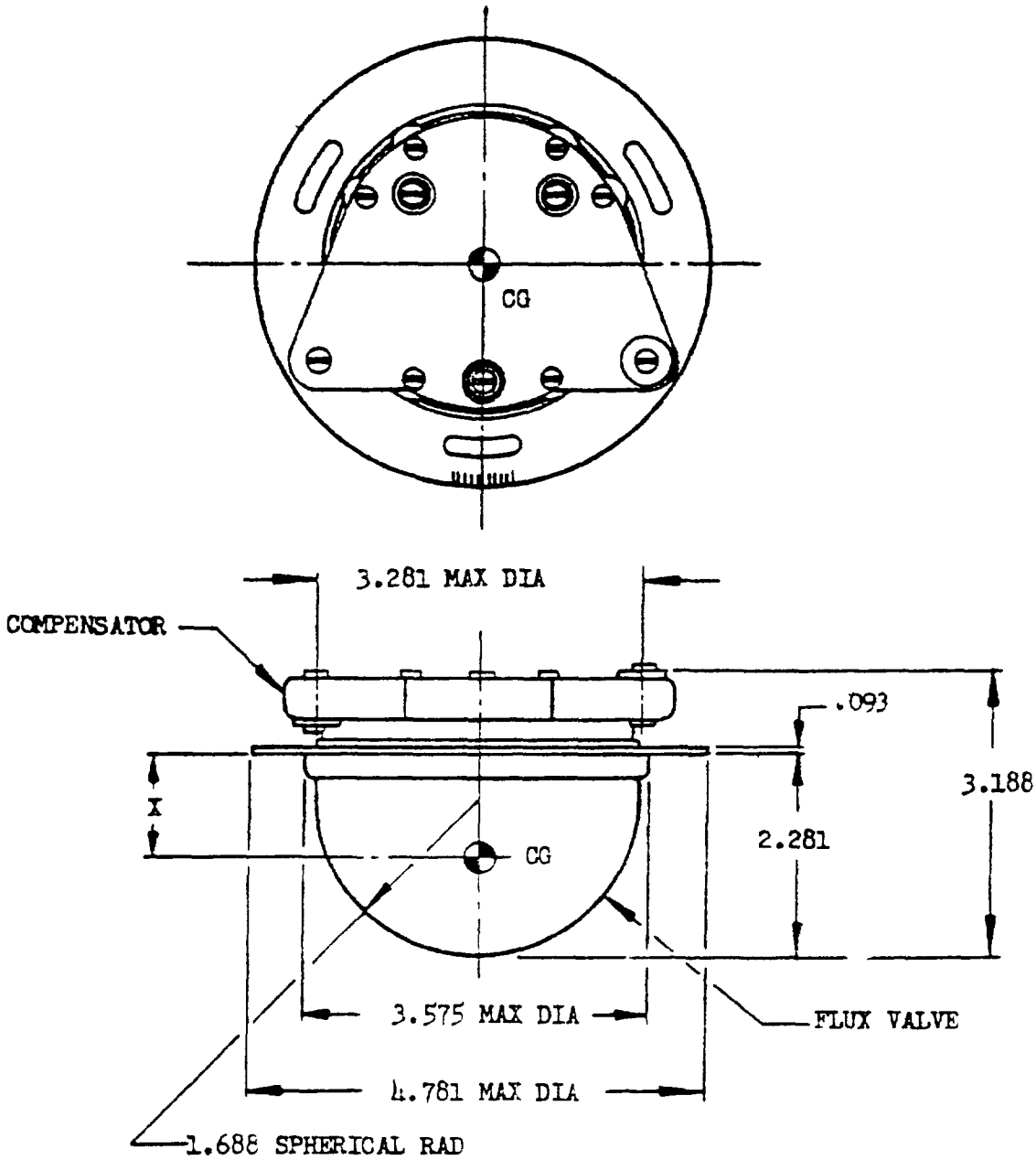
3.3.1 Metals - Metals shall be of the corrosion-resistant type or shall be suitably protected to resist corrosion during normal service life. Materials contained within hermetically sealed enclosures are considered to be suitably protected from corrosion. Requirements specified for fungicidal and corrosion protective treatment, anodizing of the aluminum-alloy parts and cadmium plating of steel parts are not applicable for parts within hermetically sealed instruments.

3.3.2 Nonmagnetic materials - Nonmagnetic materials shall be used for all parts of the compass transmitter, except where magnetic materials are essential.

3.3.3 Fungus-proof materials - Materials which are not nutrients for fungi shall be used to the greatest extent practicable. In cases where materials that are nutrient for fungi must be used, such materials shall be treated with a fungicidal agent, as approved by the procuring activity.

3.3.4 Protective treatment - When materials that are subject to corrosion in salt air or other atmospheric conditions likely to occur during service usage are used in the construction of the compass transmitter, they shall be protected

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CENTER OF GRAVITY (LEVEL POSITION)

X = 1-1/8 WITH COMPENSATOR
X = 1-5/16 WITHOUT COMPENSATOR

DIMENSIONS ARE IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCE: ±.016.

FIGURE 1. TRANSMITTER, REMOTE COMPASS, TYPE C-2

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against such corrosion in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of atmospheric conditions shall be avoided.

3.3.5 Selection of materials - Specifications and standards for all materials, parts, and Government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with ANA Bulletin No. 143, except as provided in the following paragraph.

3.3.5.1 Standard parts - Standard parts (MS, AN, or JAN) shall be used wherever they are suitable for the purpose, and shall be identified on the drawing by their part numbers. Commercial utility parts such as screws, bolts, nuts, cotter pins, etc, may be used, provided they possess suitable properties and are replaceable by the standard parts (MS, AN, or JAN) without alteration, and provided the corresponding standard part numbers are referenced in the parts list and, if practicable, on the contractor's drawings. In the event there is no suitable corresponding standard part in effect on date of invitation for bids, commercial parts may be used provided they conform to all requirements of this specification.

3.4 Design - The transmitter shall conform to figure 1. The transmitter, when properly energized and placed in the earth's magnetic field, shall be capable of providing the indicator signal transformer with the suitable direction signal for the compass system.

3.4.1 Transmitter mechanism - The transmitter mechanism shall be mounted pendulously with respect to the case.

3.4.2 Pendulous element - The earth inductor shall be so mounted on a pendulous element that it tends to be horizontal when at rest. The pendulous element shall have 30 ±2 degrees of freedom in all directions with respect to the case, and shall have a period of not less than 0.2 second, nor greater than 0.75 second.

3.4.3 Compensator - The compensator shall be made with sufficient friction to prevent any turning by vibration.

3.4.3.1 Compensation - One complete revolution of each compensating screw shall effect one half cycle of compensation. Another complete revolution in the same direction shall effect the other half cycle. Each screw shall be marked with a dot, and its surrounding plate shall also be marked with a dot. Matching of the dot on the screw with the fixed dot shall indicate

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the zero position of compensation (neutral position).

3.5 Interchangeability - All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part number shall be governed by the drawing number requirements of Specification MIL-D-70327.

3.6 Electrical characteristics -

3.6.1 Input - Normal excitation for the transmitter primary circuit shall be 23.5 ±0.5 volts, single-phase, 400 ±40 cps, alternating current.

3.6.2 Output - The signal output from the transmitter secondary circuit shall be transmitted over a 3-wire line in order that it may be connected to the stator of the signal transformer. With the transmitter under normal excitation, and placed in a normal field strength, the maximum line-to-line output voltage shall be not less than 80 millivolts. The signal transformer rotor shall be tuned for maximum 800 cps output and rotated until maximum line-to-line voltage is obtained.

3.7 Weight - The weight of the transmitter without the compensator shall not exceed 1.5 pounds. The weight of the compensator only shall not exceed 8 ounces.

3.8 Soldering - Soldering shall be in accordance with Specification MIL-S-6872.

3.9 Protection - The transmitter shall be protected to insure that sand, dust, humidity, and fungus-producing conditions in service life will not interfere with the proper operation of the transmitter.

3.10 Screw threads - Unless otherwise specified, the threads of all machine screws 0.060 inch or larger in diameter shall conform to Specification MIL-S-7742.

3.11 Finish -

3.11.1 Aluminum-alloy parts - Aluminum-alloy parts shall be covered with an anodic film conforming to Specification MIL-A-8625, except as follows.

3.11.1.1 Small holes and case inserts need not be anodized.

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3.11.1.2 Aluminum alloys which do not anodize satisfactorily shall be coated with a chemical film in accordance with Specification MIL-C-5541.

3.11.1.3 Where primary purpose of the treatment is to afford a suitable paint base, chemical treatments in accordance with Specification MIL-C-5541 may be used in lieu of anodizing.

3.11.1.4 Castings containing non-aluminum-alloy integral inserts may be treated with a chemical film in accordance with Specification MIL-C-5541 in lieu of anodizing.

3.11.1.5 When abrasion resistance is a factor, chemical films in accordance with Specification MIL-C-5541 shall not be used in lieu of anodizing.

3.12 Performance - The transmitter shall meet the performance requirements specified in section 4, when subjected to the applicable tests.

3.13 Identification of product - The transmitter assembly, subassemblies, and parts shall be marked for identification in accordance with Standard MIL-STD-130.

3.14 Workmanship -

3.14.1 General - The transmitter, including all parts and accessories, shall be constructed and finished in a workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, marking of parts and assemblies, and freedom of parts from burrs and sharp edges. Pivots, bearings, and gears shall neither bind nor shake and shall be as nearly frictionless as possible.

3.14.2 Dimensions - Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation, or performance of the compass, they shall be held or limited accordingly.

3.14.3 Screw assemblies - Assembly screws and bolts shall be tight. The word 'tight' means that the screw or bolt cannot be appreciably tightened further without damage or injury to the screw or bolt or threads.

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3.14.4 Cleaning - The transmitter shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, and other foreign material, after final assembly. Burrs and sharp edges, as well as resin flash that may crumble, shall be removed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Classification of tests - The inspection and testing of the transmitters shall be classified as follows:

(a) Preproduction tests : Preproduction tests are those tests performed after award of contract on samples that are representative in design, performance and configuration of the item which will ~~be produced on the manufacturer's production line~~.

(b) Acceptance tests: Acceptance tests are those tests performed on individual samples which have been submitted for acceptance.

4.2 Test conditions -

4.2.1 Atmospheric conditions - Unless otherwise specified, all tests required by this specification shall be made at atmospheric pressure (approximately 29.92 in. Hg) and at room temperature (approximately 77°F (25°C)). When tests are made with atmospheric pressure or room temperatures differing materially from the above values, proper allowance shall be made for the difference from the specified condition.

4.2.2 Attitude - Unless otherwise specified, the transmitter shall be tested in its normal operating position.

4.2.3 Vibration - Unless otherwise specified, the transmitter shall be subjected to sufficient vibration during test at a frequency of from 1,500 to 2,000 cpm, that a point on the transmitter will describe, in a vertical plane, a circle of not less than 0.002 nor more than 0.005 inch in diameter.

4.2.4 Supply voltage - Unless otherwise specified, all tests shall be conducted with 23.5 ±0.5 volts, single-phase, 400 ±5 cps alternating current applied to the transmitter.

4.2.5 Field strength - Unless otherwise specified, all tests shall be conducted with a horizontal field strength of 0.18 ±0.02 oersted and a vertical field strength of 0.54 ±0.02 oersted.

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4.2.6 Standard signal transformer - All tests shall be conducted with a standard signal transformer designed for this purpose. The signal transformer shall consist of a three-phase, three-wire stator surrounding a single-phase, two-wire rotor. The maximum input line-to-line impedance of the stator with normal rotor load shall be 133/81° ohms ±10 percent of the magnitude and angle, when measured at 1,000 cps with a standard laboratory instrument, such as a General Radio Type 650-A impedance bridge, or equivalent. The rotor when properly tuned to the signal shall have a minimum output voltage of 625 millivolts when the line-to-line signal input voltage to the stator is 90 ±10 millivolts, at 800 cps.

4.2.6.1 The signal transformer shall be as free as possible from inherent transmission error, in order that errors caused by its use in the compass direction signal will be kept to a minimum.

4.2.6.2 The signal transformer shall have a suitable dial, mounted concentric with its rotor shaft, and a suitable pointer on the shaft for determining angular displacement from 0 to 360 degrees of the rotor to within ±0.1 degree.

4.2.7 Turntable - All tests shall be conducted with the transmitter mounted on a turntable, accurately calibrated in increments of 15 degrees from 0 through 360 degrees.

4.2.8 Standard null meter - The standard null meter shall consist of a high-input impedance device which shall be capable of detecting the null position of the rotor of the standard signal transformer to within ±0.1 degree, when the line-to-line signal voltage of 90 ±10 millivolts is applied to the stator.

4.3 Preproduction tests -

4.3.1 Sampling instructions - The preproduction test samples shall consist of three transmitters. Samples shall be identified as required and forwarded to the testing laboratory designated in the contract or order.

4.3.2 Tests -Preproduction tests shall consist of the tests specified under Acceptance tests and, in addition, the following test.

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4.3.2.1 Special vibration test - The transmitter shall be vibrated at a frequency of 900 cpm with a total excursion of 0.125 inches in a horizontal plane, for a period of 100 hours. The transmitter shall be mounted on the vibration machine in such a manner that the zero index line on the transmitter flame is 45 degrees from the direction of the vibration machine motion. The transmitter shall then meet both the scale error test specified in 4.4.1.2 and the tilt error specified in 4.4.1.7.

4.4 Acceptance tests - Acceptance tests shall consist of Individual tests and Sampling tests.

4.4.1 Individual tests - Each transmitter shall be subjected to the following tests.

4.4.1.1 Examination of product - Each transmitter shall be inspected to determine conformance with this specification with respect to material and workmanship.

4.4.1.2 Scale error - The transmitter without compensator shall be mounted on a turntable in a normal position with standard field strength, and standard input voltage. The transmitter secondary circuit shall be connected to the stator of a standard signal transformer whose rotor is connected to a standard null meter. The transmitter and signal transformer shall be set at zero or "north" heading (rotor at null position). The transmitter shall then be turned in heading-increments of 30 through 360 degrees. At each heading, the pointer of the signal transformer shall be turned correspondingly to its null position and the reading of the pointer on the dial taken. The difference between the pointer reading and heading of the transmitter shall not exceed ±0.75 degree after the index error has been subtracted.

4.4.1.3 Neutral position test - The transmitter without compensator shall be set at the zero heading specified in 4.4.1.2, the Scale error test. The compensator, with both north-south and east-west screws set at neutral, shall then be placed in its normal position on the transmitter. The resulting shift of the null position of the signal transformer rotor shall not exceed ±1 degree. This test shall be repeated with the transmitter set on the 90-degree heading.

4.4.1.4 Compensator range test - The transmitter without compensator shall be set at the zero heading specified in 4.4.1.2, the Scale error test. The compensator with both north-south and east-west screws set at neutral shall be placed

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in its normal position on the transmitter. The north-south screw shall then be turned 180 degrees. The shift in null position of the rotor shall be between 8 and 13 degrees plus or minus, depending on the direction of rotation of the adjustment screw. The same screw shall then be turned an additional 360 degrees in the same direction. While the screw is being turned the resulting shift in null position of the rotor shall be between 8 and 13 degrees. The test shall be repeated with the transmitter set on the 90-degree heading, using the east-west screw and leaving the north-south screw at neutral. The corresponding shift in null position shall be from 8 to 13 degrees on either side of neutral position.

4.4.1.5 Interference test - The transmitter without compensator shall be set at the zero heading specified in 4.4.1.2, Scale error test. The compensator shall then be placed in its normal position on the transmitter, with the north-south screw set at neutral and the east-west screw set at 180 degrees from neutral. The resulting shift in null position of the rotor shall not exceed ±1.5 degrees. The test shall be repeated with the transmitter set on the 90-degree heading, with the east-west screw set at neutral and the north-south screw set 180 degrees from neutral position.

4.4.1.6 Field strength variation test - The transmitter shall be set up with the signal transformer as specified in 4.4.1.2, the Scale error test, and the transmitter shall be oriented to give zero index error. The horizontal field strength shall be decreased to 0.08 ±0.005 oersted. The shift in null position of the signal transformer rotor shall not exceed ±3/4 degree after the index error has been subtracted.

4.4.1.7 Tilt error test - The transmitter shall be set up with the signal transformer as specified in 4.4.1.2, the Scale error test. The transmitter shall then be tilted 15 degrees in pitch (fore end up). It shall then be rotated smoothly about a vertical axis to the 90-degree heading in a period of 20 to 30 seconds. The error between the signal transformer rotor and the transmitter shall not exceed 1-1/2 degrees after the index error has been subtracted. Similarly, the transmitter shall be rotated to and readings taken on the 180-, 270-, and 360-degree headings. The same limits shall apply. No vibration shall be applied in this test.

4.4.1.8 Signal strength test - The transmitter shall be mounted on a turntable in a normal position with a

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standard signal transformer. The transmitter and signal transformer shall be set at zero or "north" heading. Then the heading of synchronization null shall be determined by rotating the transmitter. The signal transformer heading shall be displaced from the transmitter heading and the maximum 800-cps voltage recorded. This voltage shall be not less than 600 millivolts.

4.4.2 Sampling tests - Five transmitters shall be selected at random from each quantity of 500 or fraction thereof on the order and subjected to the following tests. The five transmitters may be tested at the same time.

4.4.2.1 Low temperature test - The transmitter, while inoperative, shall be subjected to a temperature of -65°F (-54°C) for a period of 3 hours. The transmitter shall then be set into operation while at this temperature, and resubjected to the Scale error test (see 4.4.1.2), except that the tolerance shall be ±1 degree. The transmitter shall show no signs of leakage. The adjusting screws on the compensator shall operate without excess binding or excess freedom.

4.4.2.2 High temperature test - The transmitter shall be operated for a period of 3 hours at an ambient temperature of 160°F (71°C). After returning to room temperature, the transmitter shall meet the requirements of the Scale error test (see 4.4.1.2). The transmitter shall show no signs of leakage, and the adjusting screws on the compensator shall operate without excess binding or excess freedom.

4.4.2.3 Vibration endurance test - The transmitter shall be vibrated at a frequency of 2,000 cpm, so that a point on its case describes approximately a circle of a diameter of 0.018 to 0.020 inch in a plane inclined at 45 degrees to the horizontal for a period of 3 hours. The compensator and transmitter shall not be damaged, nor shall the compensator setting be changed. The transmitter shall then be resubjected to the Individual tests.

4.4.2.4 Vibration error test - The transmitter shall be vibrated as specified in 4.4.2.3, the Vibration endurance test, except that it shall be operated with the signal transformer alined on the zero-degree heading before vibration. The transmitter shall then be vibrated at frequencies of 600, 800, 1,000, etc cpm through 2,500 cpm with an amplitude of .003 to .005 inch (see 6.2.1). At each frequency, the shift in null position of the signal transformer rotor shall not exceed ±1 degree from the normal heading. The plane of the circle of vibration shall contain the east-west axis. This test shall be repeated on two other headings not less than 100 degrees apart.

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4.4.2.5 Voltage-frequency variation test - The transmitter shall be set up as specified in 4.4.1.2, the Scale error test, with the signal transformer on the zero-degree heading. The supply voltage shall then be reduced to 22 volts and the frequency to 360 cps. The shift in null position of the signal transformer rotor shall not exceed ±1/2 degree. This test shall be repeated on two other headings not less than 100 degrees apart.

4.4.2.6 High altitude-low temperature - The transmitter shall be subjected to a pressure equivalent to 70,000 ± 500 feet at a temperature of -65° ± 5°F (-54° ± 3°C) for a continuous period of 4 hours. At the end of this period there shall be no signs of leakage or any other damage which would adversely affect subsequent operation of the transmitter. After the transmitter has returned to room temperature, it shall then be subjected to the Individual tests.

4.4.3 Rejection and retest - When tests are specified on a number of items that are selected from a production run, and one or more of this number fails to meet the specified test, acceptance of all items still on hand and subsequently produced will be withheld until the extent and cause of failure is determined. For operational reasons, Individual tests may be continued pending investigation of a Sampling test failure. However, final acceptance of the items on hand and subsequently produced is contingent upon the Inspector's decision regarding the over-all conformance of the product to specification requirements. When corrective action has been accomplished, all necessary tests shall be repeated. If investigation indicates that the defects may exist in items previously accepted, full particulars concerning the defects, including recommendations for correction, will be furnished to the contracting officer.

5. PREPARATION FOR DELIVERY

5.1 Packaging -

5.1.1 Level A. - The transmitter shall be packaged in accordance with Specification MIL-P-7936. The method of preservation shall be in accordance with Specification MIL-P-116, Method 1A5, omitting preservation compound/using metal reusable containers (see 6.2).

5.1.2 Level C. - When this level is required, packaging shall be in accordance with standard commercial practice.

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5.2 Packing -

5.2.1 Levels A and B. - The instrument shall be packed in accordance with Specification MIL-P-7936. The level or levels required shall be as specified in the invitation for bid or contract.

5.2.2 Level C. - The instrument shall be packed to insure that the shipment arrives in a satisfactory condition at destination. The shipment shall conform to the applicable carriers' rules and regulations in effect at the time of shipment.

5.3 Marking - The interior and exterior containers shall be marked as specified in Specification MIL-P-7936.

5.3.1 Precautionary Marking - The following precautionary marking shall appear on two opposite sides of each interior package and shipping container whenever practicable depending on the size of the carton:

FRAGILE

DELICATE INSTRUMENTS

HANDLE WITH CARE

5.3.2 Reinspection date marking - The reinspection date marking shall be in accordance with ANA Bulletin No. 405.

6. NOTES

6.1 Intended Use - The type C-2 transmitter covered by this specification is intended for use as a component of the type J-1, J-2, J-4, N-1, S-2, MA-1 and MF-1 compass systems and automatic pilots.

6.2 Ordering data - Requisitions, contracts and orders should state the quantity desired, the levels of packaging and packing to be furnished and the name and address of the laboratory to which the preproduction samples should be submitted.

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6.3 Definition

6.3.1 Amplitude - Whenever the word "amplitude" is specified, it shall mean the extent of motion as measured from one extreme to the opposite extreme.

6.4 Provisions for preproduction tests - The manufacture of instruments on contract shall not commence until the samples submitted are pronounced satisfactory by the Bureau of Naval Weapons. When a contractor is in continuous production of the transmitter from contract to contract, submission of further preproduction samples on the subsequent contracts may be waived at the discretion of the bureau. Approval of preproduction samples or the waiving of preproduction tests does not preclude the requirement of acceptance testing.

6.4.1 It shall be understood that the transmitters supplied under contract or order shall be identical to the corresponding preproduction sample in design, construction, quality, material, workmanship and method of manufacture. Deviation from the standards of the preproduction sample shall be made only by the Bureau of Naval Weapons. Evidence of unauthorized change shall constitute cause for rejection.

NOTICE - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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